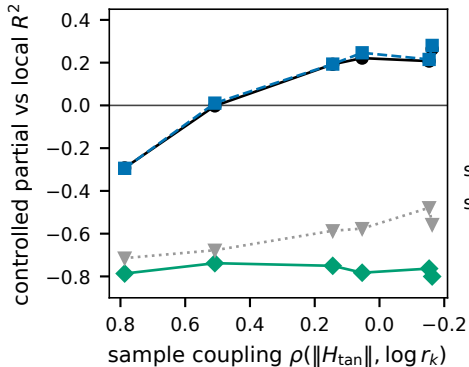
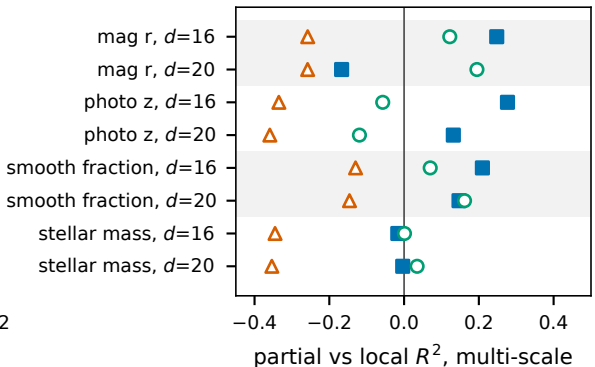


(a) known surface, intrinsic-linear label



- $\square$ exact $\|H_{\text{tan}}\|$
- - $\square$ decoder $\|H_{\text{tan}}\|$
- $\diamond$ exact $\|\langle w_N, l \rangle\|$ (here the sphere term $\sqrt{d} \|\hat{y} - b_0\|$)

(b) galaxies, decoder



- $\cdots \nabla \cdots$ exact $\|\Delta\|$ (Hessian mismatch)
- $\square$ decoder $\|H_{\text{tan}}\|$
- $\circ$ decoder $\|\langle w_N, l^S \rangle\|$ (shape)
- $\triangle$ sphere term $\sqrt{d} \|\hat{y} - b_0\|$